



Use of Web-Based Calculator for the Implementation of ACR TI-RADS Risk-Stratification System

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Abstract

In this article, we demonstrate the use of a software-based radiologist reporting tool for the implementation of American College of Radiology Thyroid Imaging, Reporting and Data System thyroid nodule risk-stratification. The technical details are described with emphasis on addressing the information security and patient privacy issues while allowing it to integrate with the electronic health record and radiology reporting dictation software. Its practical implementation is assessed in a quality improvement project in which guideline adherence and recommendation congruence were measured pre and post implementation. The descriptions of our solution and the release of the open-sourced codes may be helpful in future implementation of similar web-based calculators.

Keywords ACR TI-RADS · Web calculator · Structured reporting · Thyroid nodules · Risk-stratification systems

Introduction

Various category-based risk-stratification systems have recently been developed with the support of the American College of Radiology (ACR) to aid radiologists in the consistent interpretation and reporting of imaging examinations for various common pathologies, including the Liver Imaging, Reporting and Data System (LI-RADS) for suspected hepatocellular carcinoma in at-risk patients [1], and PI-RADS for prostate cancer [2]. These systems are useful to radiologists as they formalize and categorize imaging findings in diagnostic radiology interpretation, an otherwise subjective task, which aids in clear communication and outcome tracking.

However, widespread adoption of such systems can be plagued by intrinsic and external challenges. Some of these systems rely on cumbersome scoring systems that may make utilization tedious and the system time consuming to

use. There is usually a high learning curve, especially when an interpreting radiologist would need to be familiar with new lexicon as well as potentially complicated algorithmic charts and workflow.

An example of a risk-stratification system that may be perceived as cumbersome is the ACR Thyroid Imaging, Reporting and Data System (TI-RADS), used to stratify the risk of malignancy within a thyroid nodule based on ultrasound characteristics [2]. While the ACR TI-RADS system offers many benefits including standardization of report language and reducing inter-observer variability, it increases the tedium of reporting: five categories of ultrasound features are evaluated; each feature assigned a point value, with a total score calculated as the sum of points from each category, for up to 4 nodules. Secondly, there are various nuances in the grading system and recommendations where errors may be introduced through manual processes. For example, for a spongiform nodule (universally benign), a user is not to add additional points for other categories. In addition, a specific point value is assigned when a specific feature category is not well evaluated.

One way to mitigate these problems is the use of a software-based calculation tool. Particularly for ACR TI-RADS, the fairly unintuitive aspects of the scoring algorithm can be programmed within the tool that spare the radiologist from the burden of calculating the risk score for each nodule,

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and subsequently reducing calculation errors. While various calculators with this capability are already available online, there are further potential improvements that can be made.

In this article, we showcase a custom radiologist decision tool implemented in our institution to help radiologists utilize the ACR TI-RADS risk stratification system. A few novel aspects of this tool highlight advantages over other currently available web-based calculators including the integration with our electronic health record (EHR) system (EPIC®, *Verona, WI*) and into radiology dictation software (PowerScribe 360®, *Burlington, MA*) so that the automatically generated text is seamlessly exported to the dictation template, improving the ease of use.

Material and Methods

Software Interface Overview

Figure 1 shows the basic layout of the application. Currently, the application allows for the grading of up to 4 thyroid nodules per study, as recommended in the ACR TI-RADS white paper [3]. The main panel allows a user to indicate nodule size, nodule location, and ultrasound features within each category via radio buttons. The last category, Echogenic Foci, is multi-select allowing a user to choose all features that apply. Once the form is completed, clicking “Compute” will calculate the TI-RADS category assignment (TR1, benign; TR2, not suspicious; TR3, mildly suspicious; TR4, moderately suspicious; and TR5, highly suspicious), and generate a template output text, including the descriptive features selected and associated nodule-specific recommendations based upon the TI-RADS category. Users can adjust the ultrasound feature and recalculate if needed. Once satisfied, the user may click “Save and Transmit.” The inputs are then saved to the postgres database and transmitted to Powerscribe, a dictation software used at our institution. Within PowerScribe, our dictation software, the output text is set up to populate a merge field within the thyroid report template. The calculator can accommodate additional changes after the report was transmitted. Relaunching the calculator will open the calculator in the last saved state. Users can continue to edit the inputs and re-transmit the results to Powerscribe as long as the report has not been final signed.

Application Architecture, Integration with EHR and HIPAA Compliance

Our application was written using python programming language and Django web framework. The Django web framework uses a Model-View-Controller GUI-programming paradigm. HTML templates are used as the “view.” The “controller” comprises of python scripts that describe how

the algorithms function. The “Model” is the database where the patient data and prior inputs are stored, and with which the “controller” communicates. The database maintains the patient and per nodule characteristics data.

The software is hosted on a server located within our institution’s intranet and behind the healthcare enterprise firewall. Its security is compliant with the Health Insurance Portability and Accountability Act of 1996 (HIPAA), and therefore usable in clinical workflows. The architectural view of the setup is as shown in Fig. 2. It is connected to a postgres database to store inputs, and connected to the dictation application via RESTful api so that results can be inputted into the radiology report text.

At our institution, the EHR is integrated with the PACS system, such that the Radiology Information System (RIS) reporting palette (Epic Radiant, Verona, WI) automatically opens with opening an exam in PACS. Our TI-RADS calculator software is designed to launch via a clickable encrypted link that is accessible on the RIS reporting palette when an ultrasound examination of the thyroid is opened. The tail of the URL link is an AES128 encrypted string automatically generated by the EHR. This can be easily set up in EPIC, the EHR system of our hospital systems (Fig. 3). Figure 3a shows how the URL is set up within EPIC, and Fig. 3b shows the URL as it appears on the RIS reporting palette. After decryption, the encrypted string has the form “MRN/accession/user/institution/timeAccessed.” In this way, the calculator is opened with context specific to the patient and the study. The user is verified by cross-referencing our LDAP-based user directory for our institution to ensure only authorized providers have access to the application and time of access is recorded to allow audit capabilities. The timeAccessed is checked to ensure that an expired URL cannot be used to re-access the application beyond a 30-min window. The application design and implementation architecture were vetted by the information security committee at our institution to ensure safety and compliance with HIPAA privacy rules.

Implementation and Evaluation

Prior to implementation of this calculator, the use of the American Thyroid Association (ATA) 2015 guidelines [4] for evaluation of thyroid nodules was the most widely used guideline at our institution. When the decision was made to transition to ACR TI-RADS, the calculator was designed and released to support the change (from 2015 ATA to ACR TI-RADS). A 1-month wash in period was allowed, during which educational material was distributed and users become familiar with the new calculator. Radiologist reports for ultrasound thyroid examinations were collected prior to the wash in period (July 1, 2018 to September 15, 2018) and after (October 15, 2018 to Dec 31, 2018).

RADassist ACR TI-RADS calculator
Note: ACR TI-RADS should only be used in adult, asymptomatic patients without risk factors.

mrn: accession: date: 10-25-2021
userID:
timeStamp: 20211025181012
Site ID:

2 Nodule 1 Nodule 2 Nodule 3 Nodule 4

3

1. Location of nodule
☐ right ☒ left ☐ isthmus
☐ upper ☐ mid ☒ lower
☐ lateral ☒ medial
☒ anterior ☐ posterior

2. Size of nodule in mm (fill at least one)
 30 x 45 x 0
 mm

3. Composition
☐ Cystic
☐ Almost completely cystic
☐ Spongiform
☐ Mixed cystic and solid
☒ Almost completely solid
☐ Solid
☐ Composition obscured by calcification

4. Echogenicity of solid component
☐ Anechoic
☐ Hyperechoic
☐ Isoechoic
☒ Hypoechoic
☐ Very hypoechoic
☐ Echogenicity is indeterminant

5. Shape - Taller-than-wide?
☐ Yes
☒ No

6. Margin
☐ Smooth
☐ Ill-defined
☒ Lobulated
☐ Irregular
☐ Extra-thyroidal extension
☐ Cannot be determined

7. Echogenic foci (choose all that apply)
☒ None
☐ Large comet-tail artifacts
☐ Macrocalcification
☐ Peripheral (rim) calcifications
☐ Punctate echogenic foci

4

Findings:
 -Nodule 1: Location: lower, medial, anterior, left lobe. Size: 30 x 45 mm. Description: almost completely solid, solid component is hypoechoic, not taller-than-wide, lobulated margins, without suspicious echogenic foci. ACR TI-RADS: TR4. FNA biopsy is recommended.

Impression:
 A 45 mm left lobe nodule. TR4. FNA biopsy is recommended.
 Tessler et al. ACR Thyroid Imaging, Reporting and DataSystem (TIRADS): White Paper of the ACR TI-RADS Committee. J Am Coll Radiol. 2017 May; 14(5): 587-595.

5 Compute Reset Save and Transmit

Reference:
 1. Tessler et al. ACR Thyroid Imaging, Reporting and DataSystem (TIRADS): White Paper of the ACR TI-RADS Committee. J Am Coll Radiol. 2017 May; 14(5): 587-595.

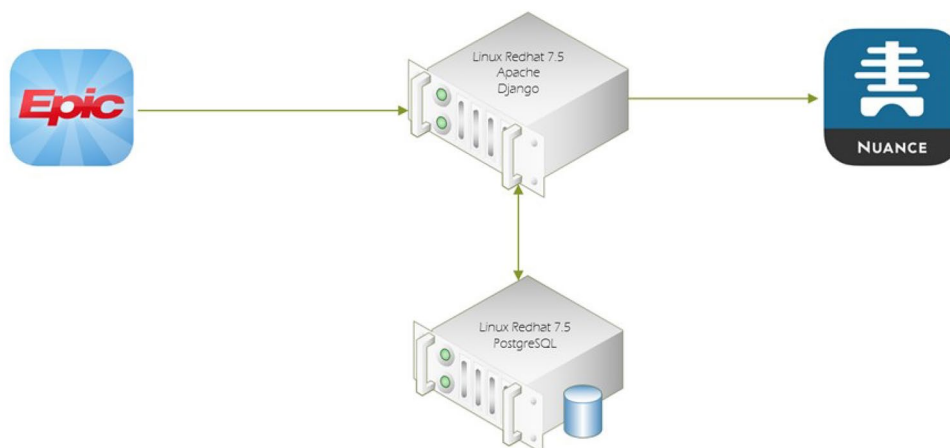
Fig. 1 General layout of the calculator. (1) Encrypted URL for transmission of context specific information such as study accession number from the EHR to the application server. (2) The application allows calculation of up to four thyroid nodules as recommended by

the ACR TI-RADS white paper. (3) Radio buttons for selection of each ultrasound features. (4) Generated report text based on the nodules' TIRADS categories. (5) Generated output text can be transmitted to the dictation report by clicking the 'save and transmit' button

For this period, a total of 1663 cases labeled US Thyroid were retrieved from our radiology report data warehouse, of which 703 cases were excluded due to duplication, erroneous orders, and cases where no thyroid nodule was reported. There were 960 cases in which thyroid nodules were reported, 508 cases pre-implementation, and 452 cases post-implementation. Among these analyzed cases, a total of 2026 thyroid nodules were described.

The parameters for evaluation include the radiologists' adherence to the agreed upon grading scheme during each time interval (2015 ATA prior to intervention, and ACR TI-RADS after the intervention). The radiologists' reports were reviewed and sorted into reports that used ATA, ACR TI-RADS, or those that does not seem to be adhering to either scoring schemes (No Scoring Scheme or NSS). In addition, the rate of reporting appropriate

Fig. 2 Architectural view of the web application setup. An encrypted URL generated from the EHR launches the calculator. The application server is connected to a postgres database. Through a RESTapi, output text is transmitted to the dictation software



recommendations (congruence) for the described nodules based on the encouraged grading scheme at the time was analyzed during each time interval. A recommendation is considered congruent if the radiologist gave a recommendation that is appropriate for the nodule grade according

to the grading scheme that was used for that study (2015 ATA prior to intervention, and ACR TI-RADS after the intervention). Statistical analyses were performed using one-tailed *t*-test, with *p*-value <0.05 considered statistically significant.

a Integration Configuration - TIRADS CALC LAUNCH [100030]

Workstation Imaging System Definitions Encryption Settings

General
Focus
Suppression

Installation Information

Type:	PACS [1]	Patient ID type:	UTSW IDX
Model record:	WEB PACS	Pat ID type dep:	
WFC installation:		User ID Format:	
Auto-start?		User ID type:	
Auto-start at:		Login domain:	
Dicom server:		Acc# transl. ext:	
Server name:		Order match ext:	
Always send study:		Context mod ext:	
Client ID:		Show wks limit:	
		OAuth2 Token:	

Installation Mnemonic Values

Name	Value
1 PATIENTOPENURL	%PATID%/%STUDIES%/%EPICUSERID:...: %/%TIMESTAMP%/utsw/
2 STUDYOPENURL	%PATID%/%STUDIES%/%EPICUSERID:...: %/%TIMESTAMP%/utsw/
3 LOGINURL	
4 LOGOUTURL	
5 CRYPTURL	https://radassist.swmed.edu/TIRADS/%CRYPTSTR:...:URL%/
6 CRYPTALGO	AES128
7 DEPRECATED_CRYPTKEY	
8 CRYPTIVLENGTH	0
9 LAUNCHTYPE	
10 PATIENTCLOSEURL	
11 SHOWTOOLBAR	
12 STUDYTITLETEXT	
13 PATIENTTITLETEXT	
14 STUDYREQACCNUM	
15 IMAGE_PLUGIN_NAME	
16 IMAGE_PLUGIN_XML_LPP	
17 DEF_IPADDR	
18 DEF_ITUSER	
19 DEF_IPADDR	

This model supports simple web-oriented PACS systems.

Several URLs may contain tokens of the form %TOKEN%. The format of these tokens will be described after the list of mnemonics. Each token may require a patient or a study context; the allowed contexts for each mnemonic are described in the mnemonic's documentation.

PATIENTOPENURL defines a URL to use for patient queries. This string may contain tokens requiring patient context, which will be replaced with appropriate values.

STUDYOPENURL defines a URL for study queries. This string may contain tokens requiring study or patient context, which will be replaced with appropriate values.

b

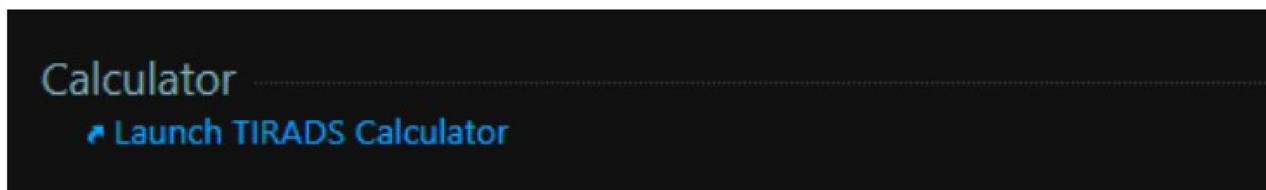


Fig. 3 Encrypted URL settings in the EHR. **a** Configuration needed to set up the URL link. **b** URL link as it shows up in the study review palette

A survey comprising of six questions were sent out to all uses of the web calculator post implementation in order to evaluate how well the calculator is received by the users. The questions are as listed on Supplemental material 1.

Software Maintenance

The maintenance of the application software and server is supported by the Information Resources team at our institution. Given the simple design of the application, it has not required any significant maintenance despite being live for more than three years.

Complete source code is open source and hosted on a Github repository <<https://github.com/UTSW-pacs/tiradsPub>>.

Results

Reports from 5 weeks prior to and 5 weeks after the transition to TI-RADS were extracted from the radiology report database. A total of 1663 studies were retrieved from the database, 644 from a county safety net hospital, and remaining 1019 cases from the University Hospital system. From studies, 704 were excluded for the following reasons: duplicate exams on a single subject, erroneous orders, and cases without thyroid nodules. A total of 959 studies remained and included 2026 individual nodules (average 2.1 nodules per subject). The patient population has an average age of 55.0

years (standard deviation: 13.8 years; age range: 18–94), and a female predominance (75.6%).

Prior to the implementation and hospital system wide switch to the TI-RADS with calculator system, 2015 ATA was the system and guideline used. In this timeframe, 53.5% used NSS, 35.3% used 2015 ATA, and 11.2% used TI-RADS with calculator. Once TI-RADS with calculator was adopted 15.0% used NSS, 0.4% used 2015 ATA, and 84.6% of reports adhered to TI-RADS with calculator. A significant decrease of 38.5% ($p < .001$) was seen in the proportion of reports that were using not using a scoring system (Fig. 4). Results were tabulated in Table 1.

In the evaluation for appropriateness of recommendations among scoring systems, when 2015 ATA was used, 70.7% of recommendations were appropriate, while 29.3% were inappropriate. When the ACR TI-RADS was used, 87.7% of recommendations were appropriate compared to only 12.3% inappropriate. A sizeable number of reports were not obviously adhering to either scoring systems. In these cases, recommendations were compared to the guidelines adopted by the department at the time (2015 ATA before implementation and ACR TIRADS after the change). In these NSS cases, 56.9% reports had appropriate recommendations, while 43.1% were inappropriate. There is a significant difference in the proportion of congruent recommendations between NSS and TI-RADS of 30.8% ($p < .001$). Similarly, there is a significant difference in the proportion of congruent recommendations between 2015 ATA and TI-RADS

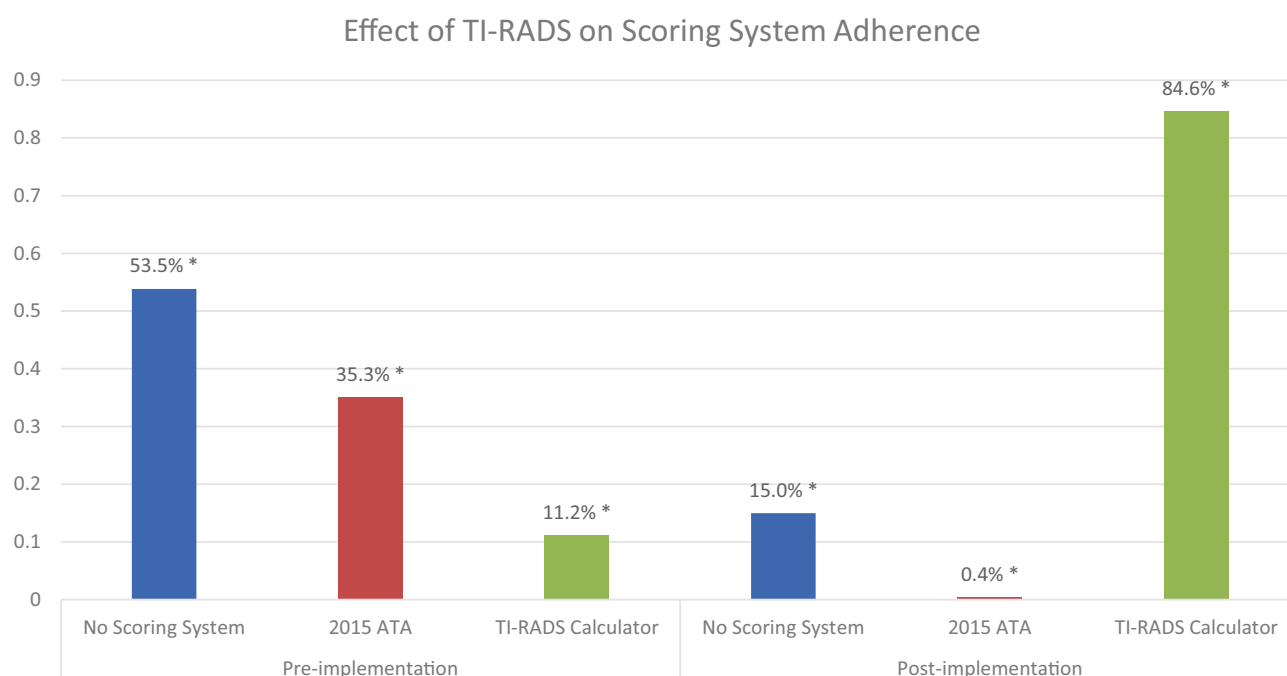


Fig. 4 Proportion of recommendations using each of the scoring systems before and after implementation of ACR TIRADS in our institution, along with the introduction of the web calculator. Asterisks mark statistically significant results

Table 1 Analyses of the recommendations for scoring system adherence and recommendation congruence

Scoring system adherence (<i>n</i> = 2026)					
Pre-implementation			Post-implementation		
No Scoring System	2015 ATA	TI-RADS calculator	No Scoring System	2015 ATA	TI-RADS calculator
53.5%	35.3%	11.2%	15.0%	0.4%	84.6%
Appropriateness of recommendations (<i>n</i> = 2026)					
Grand total		No Scoring System	2015 ATA	TI-RADS	Total
Congruent with guidelines		56.9%	70.7%	87.7%	73.3%
Different from guidelines		43.1%	29.3%	12.3%	26.7%

with calculator of 17.0% ($p < .001$). Results were shown in Fig. 5 and tabulated in Table 1.

A post implementation survey was sent out to 23 abdominal and community radiologists who read thyroid ultrasound exams at our institution. Eighteen of them replied to the survey. Results of the survey are as shown in Fig. 6. Majority of radiologists think that the calculator saves time reporting thyroid ultrasounds, helps adhere to the ACR TI-RADS scoring scheme, and increases confidence at giving follow-up/biopsy recommendations. However, most do not feel that the calculator helps to reduce misses. Most like the concept of a grading calculator and would like similar calculators built for other reporting schemes such as LIRADS for hepatocellular carcinoma and RECIST for tracking treatment response.

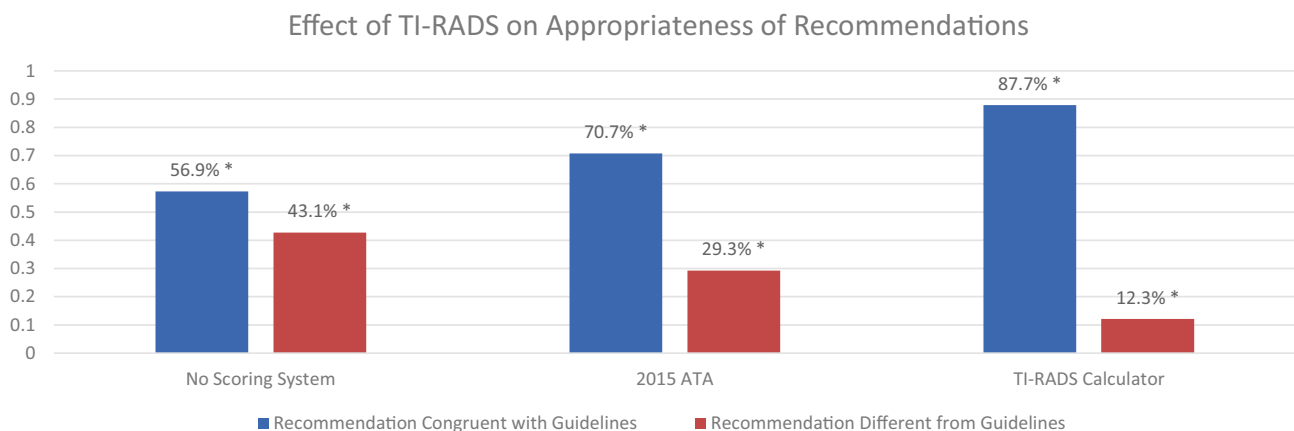
Discussion

In this article, we showcase a calculator tool for reporting nodules seen on thyroid ultrasounds utilizing TI-RADS. The motivation behind developing this custom made tool was to demonstrate the use of a software-based calculator as a possible solution to address the challenges of implementing

the growing number of risk-stratification schemes in radiology reporting. While these reporting schemes offer many benefits, there are hurdles to successful adoption, primarily due to the complexity of the scoring schemes and the high learning curve.

Our solution addressed the tedium of using one of this system (ACR TI-RADS) and showed that despite the increased complexity compared to the previously used system (2015 ATA), our results demonstrate greater adherence and recommendation congruence, and overall higher user satisfaction. There are a few additional considerations in the implementation, including the success in integration with our EHR and report dictation system while maintaining information security for protected health information (PHI). These challenges were addressed in the design of the tool including a layer of encryption and identification verification which merited the approval of the information security committee at our institution.

Attaching a database to the calculator is a design decision with additional benefits. Saving results allows for prior inputs to be pre-populated during follow up visits, requiring the user to only change features that may have changed. Since the majority of thyroid nodules do not change significantly between visits, this potentially translates to significant

**Fig. 5** Proportion of recommendations that are congruent to each of the scoring systems. Asterisks mark statistically significant results

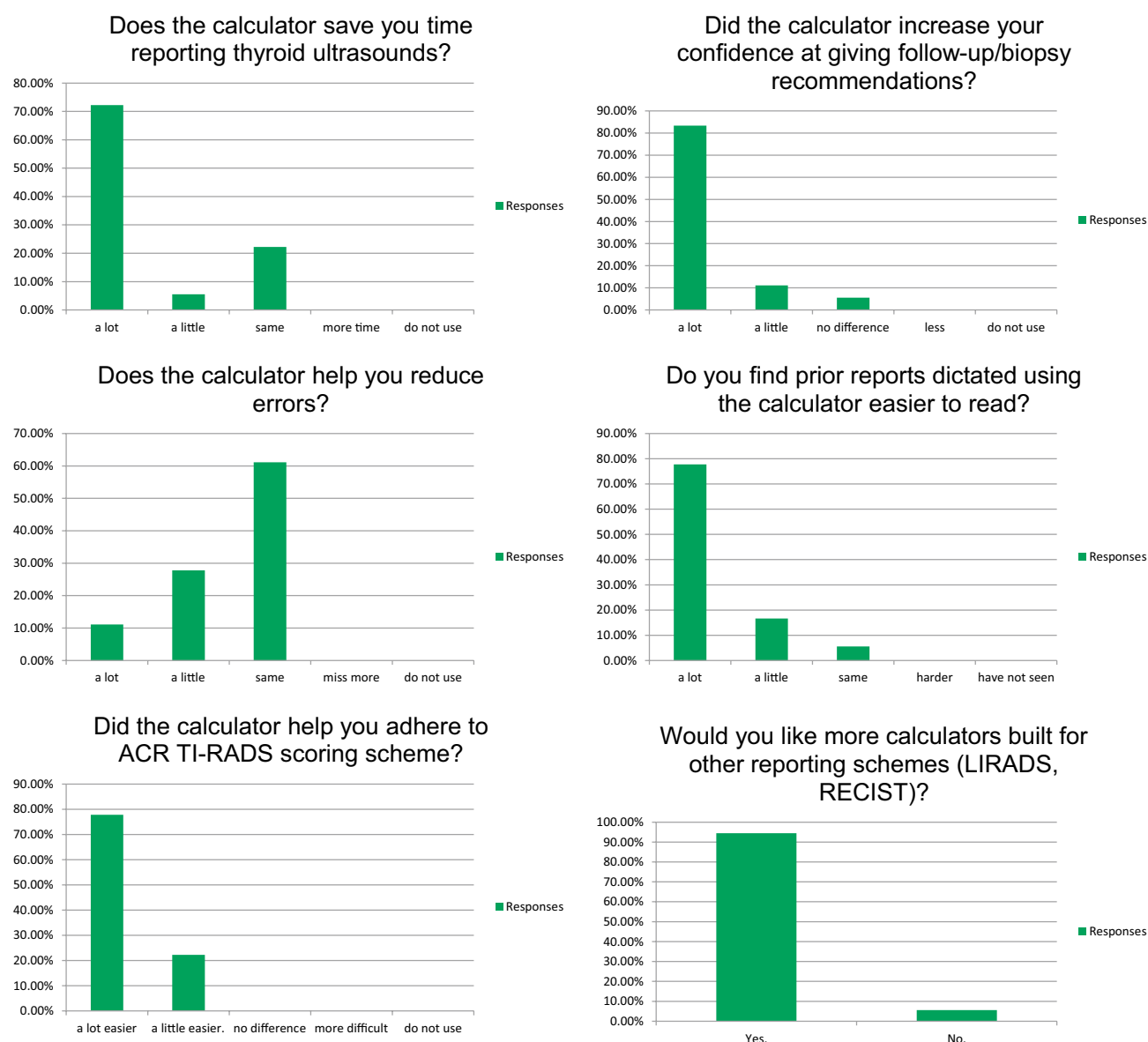


Fig. 6 Responses to the survey sent out to the users of the ACR TI-RADS web calculator

efficiencies in reporting. In our academic institution, the saving of entries also allows trainees' preliminary results to be saved, and subsequently edited prior to finalizing by the attending radiologists. Finally, the database can be used as a patient and disease registry for research, quality improvement initiatives, or volume audits.

The development of our software uncovered a few weaknesses that can be areas of further development in future iterations. First, to fully implement our solution required non-trivial familiarity with python programming language, our electronic health record functionality, and encryption. There is room to further package the software to be simpler to implement and design it to "work out of the box." A public, external facing version of the tool is

made available at <http://www.rtsfit.net/TIRADS> which has an identical computing function as our in-house tool, though lacks the integration to EHR and dictation software due to HIPAA and information security concerns. Second, grading scheme for diseases is not static and subjected to changes. As system rules are currently hardcoded into the calculator, even minor change in the guidelines (e.g. adding a new ultrasound feature for consideration) would require major reprogramming in the code and a possible delay in implementing updates clinically. Having a user-friendly graphical interface to view calculation logic and make changes to the logic may mitigate this problem and can be a potential further development direction for the future.

Conclusion

In conclusion, we have designed an integrated, web-based tool that aided our implementation of ACR TI-RADS for thyroid ultrasound reporting. We anticipate that the descriptions of our solution and the release of the open-sourced codes may be helpful in future implementation of similar web-based calculators.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10278-021-00542-2>.

Author Contribution All authors participated in project conception and writing of manuscript. Bilal Quadri analyzed the data obtained from the radiology report data warehouse and perform the statistical analyses.

Availability of Data and Material Yes. All data presented will be shared upon request.

Code Availability Complete source code is open source and hosted on a Github repository <<https://github.com/UTSW-pacs/tiradsPub>>.

Declarations

Conflict of Interest The authors declare no competing interests.

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